

Name: _____

Instructions: Please answer all of the following 8 questions. I have provided the MIPS reference sheet for you.

Notes:

- Each question will be submitted separately
- Take your time
- Show your work

Name: _____

1. (10%) MIPS Basics and Definitions
 - A. How many registers do we have in MIPS (only worry about the version of MIPS that we have been using in class)?
 - B. How many bits do we need to represent the above amount of registers?
 - C. How many bits would we need to represent 16 registers?
 - D. What are the two types of memory we have discussed and how are they different?
 - E. Name three of the bitwise operations we used in the homeworks and explain what they do

Name: _____

2. (20%) Integer Representation and MIPS to Hex
 - A. Convert the following 2 binary numbers to hexadecimal (hex)
 - a. 0101 0011 1010 1011
 - b. 1101 1000 1110 1001

 - B. Convert the following MIPS to Hex
 - a. sub \$s1, \$t2, \$a0
 - b. addiu \$s3, \$t4, \$v1

 - C. Convert the following Hex to MIPS
 - a. 0x004D8824
 - b. 0xAE08000C

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3. (20%) C to MIPS. Assume that the variables f, g, h, i, j, and k are assigned to registers \$s0, \$s1, \$s2, \$s3, \$s4, \$s5 respectively. Assume that the base address of the array A is in register \$s6.

A. What is the corresponding MIPS assembly code for the C statement

$g = (k \mid 6) - (i + j);$ // $\mid$ stands for logical OR, $\&$ is logical AND

B. What is the corresponding MIPS assembly code for the C statement (Remember to move the address then use lw or sw)

$A[j] = A[2] + (i - j);$

C. What is the corresponding MIPS assembly code for the C statement

$i = A[i-j] + A[k-4];$

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4. (20%) Logical statements and branching

- A. What happens to a number when we shift left by 2?
- B. What happens to a number when we shift right by 5?
- C. What happens to a number when we shift right by n?
- D. The register \$t0 has the value 0x80000001 in it (this is reset at the beginning of each part) (Answer in hex or binary is all I need) (These are unsigned)
 - a. What is the value of \$t2 after the following line?
srl \$t2, \$t0, 8
 - b. What is the value of \$t2 after the following lines?
sll \$t0, \$t0, 6
sli \$t2, \$t0, 128
 - c. What is the value of \$t2 after the following lines?
sll \$t0, \$t0, 4
addi \$t1, \$zero, 16
beq \$t0, \$t1, Else
andi \$t2, \$t0, 1
j Done
Else: ori \$t2, \$t0, 1
Done:

Name: _____

5. (20%) C to MIPS

A. Convert the following C code to MIPS (use registers from problem 3)

```
while(k < 5)
{
    i--;
    j++;
    k += 2;
}
```

B.

```
test_example(int a)
{
    int i = a * 4;
    for(int j = 0; j < i; j++)
    {
        print(i)
    }
    return a;
}
```

```
test_example(3);
```

(For this one write the code to call it and the code for the function)

Name: _____

6. (15%) CPI

A. What does CPI stand for?

B. Given the following tables, compute the CPI for programs 1 and 2.

Instruction	Cycles
A	1
B	2
C	7
D	3
E	1
F	6

Program	Instruction Count					
	A	B	C	D	E	F
1	2	4	0	3	9	1
2	1	3	7	1	3	2

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7. (15%) Floating Point

- A. What is the difference between Single and Double precision?

- B. What is the Bias for each precision?

- C. What is the Floating Point representation of (in single precision)(only need 5 decimals of precision):
 - a. .625
 - b. -.20625
 - c. .04325

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8. (10%) Cache and Unrolling

A. Draw a diagram showing how the CPU, L1, and L2 cache interact (either draw it on paper and submit a picture or use paint on your computer)

B. Unroll the following loop 2 times and 4 times (show both new for loops):

```
for(int i = 0; i < 20; i++){  
    print(i);  
}
```